

NoVectDB: Typed Relational Retrieval as a Complement to Vector Search

CookiX: A Reference Implementation, and What It Does and Does Not Buy

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Abstract

Retrieval-augmented generation grounds language models in external knowledge predominantly through vector similarity. That works well for topical lookup but degrades on *relational* questions — ones whose answer depends on a typed, directed connection (*prevents, causes, depends on*) rather than on proximity — and it returns a scalar distance in place of a justification. We describe **NoVectDB**, a design position rather than a replacement: retrieval substrates should carry typed directed edges alongside vectors, and should return the *path* that justifies an answer. We define the Knowledge Object $K = (V, E, \mathcal{T}, \mathcal{S})$, give a composite distance over the resulting graph in which every term is ablatable, and present **CookiX**, an open-source single-node reference implementation in pure Python.

We report what we measured, including where it loses. On 2WikiMultiHopQA [11], typed multi-hop traversal reaches hits@10 0.580 against Okapi BM25’s [12] 0.386 (+50% relative) *under oracle entity linking*, and recovers the gold relation chain on most answered questions. Without the oracle, a surface-form linker achieves 59.5% link accuracy and end-to-end hits@10 0.378 — *parity with, not victory over*, BM25’s 0.386. Entity linking and triple extraction, not traversal, are the binding constraints: measured per-edge extraction recall is 0.444, which compounds to $\approx 4\%$ survival for a four-hop chain.

Neither exploratory layer earns its place on our evidence. The topological signature ($\mathcal{T}$) changes no retrieval metric we measure. For the sheaf layer ($\mathcal{S}$) we introduce a discriminative test — does the composition residual rank a gold evidence chain below a corrupted one on held-out data? — and report a **negative result** across nine configurations: learned orthogonal restriction maps do not separate chains better than an identity-map control by more than sampling noise. We identify the mechanism: the residual is blind to which of several genuine paths connects two endpoints, and that is exactly the discrimination retrieval re-ranking requires. We also state explicitly that no dense vector database has yet been benchmarked against CookiX, and that doing so is the most important outstanding experiment.

Keywords: knowledge graphs, multi-hop retrieval, retrieval-augmented generation, explainable retrieval, sheaf theory, topological data analysis, negative results

1 Introduction

The dominant pattern for grounding language models in external knowledge is retrieval-augmented generation [1], which relies largely on embedding text into $\mathbb{R}^n$ and retrieving neighbours by cosine or Euclidean distance. This is effective for topical retrieval. It is weaker for a specific class of question, and it is that class we address.

1.1 Motivating example: the umbrella problem

Consider five concepts: *raincoat*, *rain*, *water*, *umbrella*, *storm*. A sentence transformer maps them to points that cluster tightly. The query “what prevents rain from reaching the coat?” retrieves all five with similar scores. The correct answer, *umbrella*, depends on a directed edge

$$umbrella \xrightarrow{\text{prevents}} rain \xrightarrow{\text{causes}} wet_coat,$$

and no ranking by a *symmetric distance* can encode the verb “prevents” (Proposition 2.4).

The scope of that statement matters. It is a limitation of *distance-ranked* retrieval, not of vector representations as such. Knowledge-graph embedding methods with *asymmetric scoring functions* — TransE [8], ComplEx [9], RotatE [10] — were designed precisely to represent directed relations in vector space, and nothing here refutes them. The gap we exploit is between typed traversal and *cosine-similarity retrieval over text embeddings*, which is what production RAG stacks deploy.

1.2 Contributions

1. We formalise the NoVectDB data model (Section 3): Knowledge Objects, the graph substrate, and a composite distance whose every term is ablatable.
2. We give the retrieval pipeline and its complexity (Section 7), as implemented.
3. We report reproducible benchmarks (Section 8) on a synthetic corpus and on 2WikiMultiHopQA, separating the *engine under oracle linking* from the *end-to-end system*, and reporting the latter as parity rather than a win.
4. We quantify extraction as the binding constraint on multi-hop accuracy (Section 8.4), including the compounding model.
5. We report a **negative result** for the sheaf layer (Section 8.5), with a structural diagnosis of why it fails at the task retrieval needs.

2 Where distance-ranked retrieval struggles

2.1 Concentration of measure, and how much it actually matters

Theorem 2.1 (Distance concentration [2]). *Let $X_1, \dots, X_m$ be i.i.d. random vectors drawn uniformly from the unit ball $B^n(0, 1) \subset \mathbb{R}^n$. Then for $i \neq j$,*

$$\Pr \left[\left| \|X_i - X_j\| - \sqrt{2} \right| > \varepsilon \right] \leq 2 \exp(-c n \varepsilon^4),$$

for an absolute constant $c > 0$.

Remark 2.2 (Scope). This should not be read as a practical indictment of vector retrieval. The theorem concerns vectors drawn *uniformly from a ball*; learned text embeddings are strongly anisotropic and concentrate near low-dimensional manifolds, which is why cosine retrieval remains discriminative at $n = 768\text{--}3072$ in practice. Concentration of measure is an asymptotic caution about isotropic data, not the failure mode practitioners observe. Our argument rests instead on Section 2.2, which is independent of it.

2.2 The semantic gap: a symmetric distance cannot encode an asymmetric relation

Definition 2.3 (Semantic gap). Let $\phi : \mathcal{D} \rightarrow \mathbb{R}^n$ be an embedding of a corpus $\mathcal{D}$, and let ρ be a target relation (e.g. *causes*, *contradicts*). The embedding exhibits a *semantic gap* with respect to ρ if there exist $a, b, c \in \mathcal{D}$ with

$$\rho(a, b) = \text{true}, \quad \rho(a, c) = \text{false}, \quad \|\phi(a) - \phi(b)\| \geq \|\phi(a) - \phi(c)\|.$$

Proposition 2.4. Let d be any metric on $\mathbb{R}^n$ and let retrieval rank candidates by $d(\phi(a), \cdot)$. For any non-symmetric relation ρ there exist corpora on which this ranking exhibits a semantic gap.

Proof. A non-symmetric ρ distinguishes the ordered pairs (a, b) and (b, a) , whereas any metric satisfies $d(x, y) = d(y, x)$. A ranking that is a function of d alone therefore assigns the same score to both orderings and cannot separate them. $\square$

Remark 2.5 (What this does and does not cover). Proposition 2.4 is a statement about *metric-ranked* retrieval. An asymmetric scoring function over embeddings is not a metric and escapes the argument entirely; the knowledge-graph embedding literature [8, 9, 10] is built on exactly that escape. The claim we make is narrower and, we think, still worth making: the deployed default — rank text embeddings by cosine — cannot represent direction.

2.3 Opacity

Vector retrieval returns a scalar. There is no path and no typed connection, so when a RAG system answers incorrectly, attributing the error to a retrieval decision requires inspecting inner products. Returning the justifying path is the capability we consider most defensible, and Section 8 reports it as `path_match`.

3 The NoVectDB data model

NoVectDB stands for *Not Only Vector Database*, after NoSQL: vectors are one tool among several.

Definition 3.1 (Knowledge Object). A Knowledge Object is a quadruple $K = (V, E, \mathcal{T}, \mathcal{S})$:

- $V \in \mathbb{R}^n$ (optional): an embedding or content signature.
- $E = \{(r_i, K_i, w_i)\}$: typed, directed, weighted edges, with r_i drawn from an extensible vocabulary $\mathcal{R} = \{\textit{causes}, \textit{is_a}, \textit{part_of}, \textit{prevents}, \textit{contradicts}, \dots\}$ carrying algebraic properties (symmetry, transitivity, inverse).
- $\mathcal{T} \in \mathbb{R}^m$ (optional, *no measured benefit*): a topological signature from persistent homology of the local neighbourhood.
- $\mathcal{S}$ (optional, *negative result*): a sheaf stalk with linear restriction maps per edge.

V is optional: pure topological–relational storage is a valid configuration.

Definition 3.2 (Composite distance). For Knowledge Objects K_a, K_b ,

$$d(K_a, K_b) = \alpha \cdot d_{\text{geo}}(a, b) + \beta \cdot (1 - \text{TVS}(\mathcal{T}_a, \mathcal{T}_b)) + \gamma \cdot \|\mathcal{S}_a \circ_{\pi} \mathcal{S}_b\|, \quad (1)$$

with $\alpha + \beta + \gamma = 1$, where d_{geo} is the minimum-weight path distance, TVS compares persistence signatures, and the third term is the sheaf composition residual along path π .

Setting $\beta = \gamma = 0$ recovers pure typed traversal. Every result in Section 8 is reported per ablation, and on our measurements $\beta, \gamma > 0$ never helps. We regard that as the point of building the distance this way: the exotic terms are hypotheses held where they can be switched off and measured.

4 Exploratory layer I: topological signatures

Definition 4.1 (Topological signature). For a Knowledge Object K with r -hop neighbourhood $N_r(K)$, build the Vietoris–Rips complex on the shortest-path distance matrix using edge weights as filtration values, and set $\mathcal{T}(K) = \text{Vectorise}(\text{Barcode}(\text{VR}(N_r(K))))$.

Definition 4.2 (Topological vector similarity). $\text{TVS}(\mathcal{T}_a, \mathcal{T}_b) = \exp(-\lambda W_2(\text{Dgm}_a, \text{Dgm}_b))$ for bandwidth $\lambda > 0$ and 2-Wasserstein distance W_2 [3].

Remark 4.3 (Status: no measured benefit). Across every benchmark in Section 8, enabling $\mathcal{T}$ changes no retrieval metric outside noise. The layer remains in the implementation because it is ablatable and costs nothing switched off, not because we have evidence for it. The hypothesis — that the *shape* of a concept’s neighbourhood carries retrieval signal — is not refuted by this, but it is not supported either, and we make no claim for it.

5 Exploratory layer II: sheaf composition

Definition 5.1 (Cellular sheaf [4, 5]). A cellular sheaf $\mathcal{F}$ on a directed graph $G = (V, E)$ assigns a vector space (*stalk*) $\mathcal{F}(v) \cong \mathbb{R}^d$ to each vertex and a linear *restriction map* to each edge. In CookiX a stalk encodes an object’s local semantic frame and a typed edge’s restriction map encodes how meaning transforms across that relation.

Definition 5.2 (Composition residual). For a path $\pi = (v_0, e_1, v_1, \dots, e_k, v_k)$, the composed map is $S_\pi = \mathcal{F}_{e_k} \circ \dots \circ \mathcal{F}_{e_1}$ and the residual is

$$\text{res}(\pi) = \|S_\pi(x_{v_0}) - x_{v_k}\|. \quad (2)$$

The intended reading is that a low residual indicates a coherent chain.

Remark 5.3 (Structural blindness). Equation (2) reads only the source stalk, the relation sequence, and the target stalk. Intermediate vertices never enter it. The residual therefore *cannot* distinguish two paths that share endpoints and relation sequence, and as Section 8.5 shows empirically, it does not distinguish paths that share endpoints with a *different* relation sequence either. This is a property of the definition, not of a particular fit, and it is the reason the layer fails at re-ranking.

6 Retrieval-theoretic analysis

Definition 6.1 (Relational query). A relational query $q = (K_{\text{start}}, \rho_1, \dots, \rho_k)$ seeks objects reachable from K_{start} along a chain of typed relations.

Proposition 6.2 (Conditional advantage). *Let $\text{Prec}_V(q)$ be the precision@k of metric-ranked vector retrieval and $\text{Prec}_N(q)$ that of typed traversal, for a query of hop count h . Suppose each required edge is present and correctly typed in the graph independently with probability p , and the head entity is correctly identified with probability ℓ . Then*

$$\text{Prec}_N(q) \lesssim \ell \cdot p^h, \quad (3)$$

and traversal outperforms metric-ranked retrieval only when $\ell p^h > \text{Prec}_V(q)$.

Sketch. Traversal recovers the target only if the anchor is correct and every hop’s edge is present and correctly typed; independence gives the product. Correct traversal is precision 1.0 for single-hop typed lookup, so the bound is tight at $h = 1, \ell = p = 1$. $\square$

Remark 6.3 (The condition is not satisfied in the open-domain setting). Equation (3) is the honest centre of this paper, and it cuts against us. At our measured values — $p \approx 0.444$ from Section 8.4 and $\ell \approx 0.595$ from Section 8.3 — two-hop survival is $0.595 \times 0.444^2 \approx 0.117$, and the inequality fails against a BM25 baseline at 0.386. This is not a discrepancy to be explained away: it is exactly what the end-to-end measurement shows, where CookiX reaches parity rather than beating BM25. Under oracle linking ($\ell = 1$) with a gold-triple graph ($p = 1$) the same inequality predicts the advantage we do observe. The advantage is real and the condition for it is specific.

Corollary 6.4 (Dimension independence). *Because d_{geo} is defined on a discrete graph and TVS derives from topological invariants, neither depends on an ambient embedding dimension.*

Remark 6.5. This is a categorical difference, not a demonstrated benefit; given Remark 2.2 it should not be read as an advantage over deployed vector retrieval.

7 CookiX: architecture

CookiX is a document-oriented embedded library, in pure Python (3.10+) with NumPy and NetworkX as its only core dependencies.

- **Ingestor.** Accepts structured documents, or free text via a swappable extractor: a deterministic rule-based keyword extractor, or an LLM-backed extractor through an API. Extraction quality is the ceiling on everything downstream (Section 8.4), which is why it is a first-class component.
- **Storage.** Three interchangeable backends behind one interface, held to a shared parity test battery: an in-memory NetworkX store; a crash-safe durable store in pure Python (write-ahead log with per-record CRC for torn-write tolerance, atomic snapshots via temp-file rename, single-`fsync` transactions, re-entrant write lock); and an embedded Kùzu property graph.
- **TopoIndex.** Approximate nearest-neighbour search over persistence signatures by cosine LSH, with an exact fallback and a built-in recall measure.
- **Query engine.** The pipeline of Algorithm 1. Geodesic search is a settle-once Dijkstra — a node is never re-expanded once its minimum-cost path is fixed — with early exit when a specific target is requested.
- **Interfaces.** An embedded Python API; an HTTP server with API-key auth, role separation, rate limiting, resource caps, read-only mode, health probes and Prometheus metrics; and a dependency-free typed client over a versioned wire API.

Proposition 7.1 (Query complexity). *The pipeline runs in $O(|V_{\text{local}}| + |E_{\text{local}}| \log |E_{\text{local}}|)$, where the local subscripts denote the explored frontier — bounded by degree and hop limit, not by total object count.*

Algorithm 1 NoVectDB retrieval pipeline

1. $I \leftarrow \text{INTENTPARSE}(q)$ (anchor / relation / target)
 2. $S_0 \leftarrow \text{DETERMINISTICLOOKUP}(G, I)$ (exact typed-edge match; inverse relations resolved virtually)
 3. **if** a bare relation query **then return** $S_0[:k]$
 4. $S_1 \leftarrow \text{GEODESICSEARCH}(G, I.\text{anchor}, h_{\max})$ (settle-once Dijkstra, early exit on target)
 5. $S_2 \leftarrow \text{RANK}(S_1, d)$ (Eq. (1), terms enabled per ablation mode)
 6. **return** $S_2[:k]$
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8 Experiments

Every number in this section is produced by a named command in this repository, with a fixed seed. Ablation modes are reported separately throughout.

8.1 Synthetic relational corpus

`cookix eval -seed 0 -worlds 40 -k 5`. The corpus is built so that each entity’s text describes the entity itself, never its relations, and entities within a world share a topical adjective — so a content retriever genuinely retrieves the right neighbourhood, while the relational answer lives only in the typed edges. 240 documents, 160 queries (single-hop forward, single-hop inverse, multi-hop, contradiction).

retriever	hits@1	hits@5	P@5	R@5	MRR	path_acc
random	0.006	0.019	0.004	0.019	0.010	0.000
lexical-tfidf	0.250	0.750	0.150	0.750	0.375	0.000
cookix-graph	1.000	1.000	0.200	1.000	1.000	1.000
cookix-topo	1.000	1.000	0.200	1.000	1.000	1.000
cookix-sheaf	1.000	1.000	0.200	1.000	1.000	1.000
cookix-reasoning	1.000	1.000	0.200	1.000	1.000	1.000

Table 1: Synthetic corpus, seed 0. CookiX saturates, so the exotic layers have no headroom to demonstrate anything and we claim nothing from their tying.

Remark 8.1 (What this table is worth). This is a corpus we designed, and saturation at 1.000 is evidence that the implementation does what it is built to do — not evidence about the world. The informative part is the baseline: the lexical retriever reaches $R@5 = 0.75$ while scoring $\text{hits}@1 = 0.25$ and $\text{path_acc} = 0$. Content similarity finds the right neighbourhood, cannot pick the relationally correct entity out of it, and structurally cannot produce a path.

8.2 2WikiMultiHopQA under oracle entity linking

`cookix eval -dataset 2wiki -path dev.json -k 10`. One global knowledge graph is built from every example’s gold evidence triples, so traversal faces real distractor edges from thousands of

other questions. Baseline is Okapi BM25 over the same paragraphs. First 2,000 dev examples, 1,802 evaluable.

retriever	hits@10	MRR	path_match
BM25 (strong lexical)	0.386	0.239	n/a
cookix-graph	0.580	0.282	0.579
cookix-reasoning	0.580	0.283	0.579

Table 2: Oracle entity linking: the question’s head entity is given, so this measures the reasoning engine in isolation. `path_match` is unavailable for BM25 by construction — it returns passages, not paths.

This is the strongest evidence for the typed-traversal core, in a setting standard in KG-QA. It is *not* an end-to-end open-domain number, and we do not present it as one. Note also that **reasoning** ties **graph**: the re-ranking layers reorder candidates without changing which answers are reachable.

8.3 Dropping the oracle: the end-to-end result

`cookix eval -dataset 2wiki -no-oracle -linker {surface,bm25}.`

anchor source	link accuracy	CookiX hits@10	MRR
oracle (given)	100%	0.580	0.282
surface linker	59.5%	0.378	0.183
BM25 linker (context match)	50.1%	0.340	0.168
BM25 <i>retriever</i> (no linking needed)	—	0.386	0.239

Table 3: End-to-end, same 2,000 examples. A correct-signal linker lifts CookiX from below BM25 to parity. It is not a win.

Stated without spin: the engine dominates when linked correctly; matching entity names against the question beats matching paragraph content (50.1% $\rightarrow$ 59.5% link accuracy, 0.340 $\rightarrow$ 0.378 hits@10); and that lifts CookiX from *losing* to *parity* (0.378 vs 0.386) while still returning a reasoning path. Roughly 40% of questions still begin from the wrong anchor and cannot recover. Clearing $\approx$ 70% link accuracy is what would flip the comparison. An LLM-assisted linker is implemented and unit-tested; we report no number for it because we have not run the keyed benchmark.

8.4 Extraction is the binding constraint

`cookix eval -extraction.` Rule-based extractor against 16 hand-annotated sentences (18 gold triples):

extractor	precision	recall	F1	relation accuracy
rule-based	0.615	0.444	0.516	1.000

Table 4: Relation *typing* is perfectly reliable — whenever the extractor spans the right entities it labels the relation correctly. Free-text *coverage* is the bottleneck.

Projecting per-edge recall through Eq. (3) with $\ell = 1$:

	1-hop	2-hop	3-hop	4-hop
rule-based	0.444	0.198	0.088	0.039

Table 5: A four-hop chain survives about 4% of the time.

The misses are the realistic cases: out-of-vocabulary synonyms (“leads to”, “ward off”), sentences carrying two relations where a keyword splitter emits only the first, and preposition boundaries. How much of this ceiling an LLM extractor buys back is open; we have not measured it.

8.5 A negative result for the sheaf layer

`cookix eval -sheaf-probe -path dev.json -dim 32.`

An absolute residual is not a meaningful test of this layer. Fitting restriction maps on data generated to be sheaf-consistent and observing that the residual drops establishes only that orthogonal Procrustes recovers a linear map when a linear map produced the data. The question that bears on retrieval is discriminative.

Design. On held-out 2WikiMultiHopQA chains, we ask whether the residual ranks a gold evidence chain below a corrupted one. Stalks are derived from entity *text* (TF-IDF, randomly projected, L2-normalised) rather than from an entity-id hash, since a hash-seeded stalk carries no semantics and no result computed against it would mean anything. Corruptions vary only what Eq. (2) can see: a swapped relation label, a swapped source or target, a shuffled relation order, and a *detour* — a genuine alternative path between the same endpoints. Three map families are scored:

- *placeholder* — the shipped random orthogonal maps. The null hypothesis.
- *identity* — every map is I , collapsing the residual to endpoint distance $\|x_a - x_b\|$ and ignoring the relation chain entirely. This is the decisive control: it asks whether the sheaf machinery does anything that embedding distance does not.
- *learned* — orthogonal Procrustes fitted on training-split edges only, with every test-chain triple withheld.

The decision rule was fixed before running: *learned* must beat *both* controls by more than the sampling-noise floor $1/\sqrt{n}$. Beating the placeholder alone would show only that fitting beats not fitting.

Result. 12,576 dev examples yield 6,459 connected chains; 1,938 are held out and scored, with 26,126 training edges over 34 relations (median 92 edges/relation against $\text{dim} = 32$, so the fit is not underdetermined). Noise floor 0.023.

Across nine configurations ($\text{dim} \in \{8, 16, 32\}$, three seeds) the *learned*–*identity* gap ranges from +0.004 to +0.018 and is **below the noise floor in all nine**. We therefore report a negative result: on real multi-hop data, learned linear restriction maps over lexical stalks do not discriminate coherent from corrupted reasoning chains better than endpoint distance.

maps	pooled AUC	detour	rel. swap	shuffle	src. swap	tail swap
placeholder	0.509	0.498	0.504	0.524	0.502	0.508
identity (control)	0.530	0.463	0.500	0.523	0.563	0.536
learned	0.552	0.481	0.550	0.530	0.565	0.561

Table 6: AUC = $\Pr(\text{gold residual} < \text{corrupted residual})$; 0.5 is chance. $\text{learned} - \text{identity} = 0.022$, below the 0.023 noise floor.

Diagnosis. The *detour* column is 0.481 — chance. The residual cannot tell a gold chain from another genuine path between the same endpoints. This explains mechanically why $\gamma > 0$ never improves retrieval in our benchmarks: retrieval candidates *are* all real paths in the graph, i.e. all detours, which is precisely the discrimination the residual lacks (cf. Remark 5.3). The layer was moderately effective in a controlled setting at detecting wrong relation labels and wrong endpoints — a *verification* question, not a retrieval one.

What remains open. Stalks here are lexical projections, not neural sentence embeddings, and the maps are linear and closed-form rather than jointly learned as in neural sheaf diffusion [6]. Those are the substantive untried variables. We claim only what we tested: the linear-maps-over-lexical-stalks formulation does not work, and the burden of proof now sits with the neural variant.

8.6 Performance and scale

`cookix eval -perf`, 240 documents, 160 queries $\times$ 3 repeats, single-threaded: median 0.06 ms graph-only, 0.07 ms with all layers ($\approx 13,000$ vs $\approx 6,500$ q/s). `cookix eval -scale`, random typed graph, out-degree 4, four-hop traversals:

objects	ingest	peak memory	median query	p95
1,000	0.08 s	3 MB	1.5 ms	2.8 ms
10,000	0.9 s	30 MB	2.0 ms	3.1 ms
50,000	6.4 s	155 MB	2.2 ms	6.1 ms

Table 7: Latency stays near-flat across a $50\times$ increase in graph size, because traversal cost is bounded by the local reachable frontier rather than total object count. Single-threaded pure Python on one machine; the largest graph measured is 5×10^4 objects.

`cookix loadtest`: 8 concurrent clients against the real HTTP server over a 10,000-object graph sustained ≈ 130 req/s with 0 errors in 2,613 requests (median 59 ms, p99 102 ms) and non-monotonic memory. That throughput is the honest ceiling of a single synchronous Python process; multi-worker deployment is untested.

8.7 No dense-retrieval baseline has been run

We state this explicitly rather than leaving it to be inferred. CookiX has been compared against a random floor, TF-IDF cosine, and Okapi BM25. It has **not** been compared against a dense embedding retriever, Chroma, Pinecone, Weaviate, or GraphRAG. TF-IDF stands in for the vector-similarity family on the grounds that a dense embedder plugs into the same interface and behaves the same way with respect to *relations* — it retrieves by topical proximity, not traversal — but that

is an argument, not a measurement. Running a dense baseline is the most important outstanding experiment in this line of work.

9 Discussion

When to use what. For fuzzy semantic search over unstructured text, use a vector database. For a production graph database at scale, use Neo4j or Kùzu. For LLM-grounded summarisation over a corpus graph, use GraphRAG [7]. The niche we can defend is narrow: *embedded, explainable, typed multi-hop retrieval where the justifying path is part of the answer*, over data that is already relational or can be made so reliably.

Multiplicative fragility. Equation (3) deserves emphasis as a design property, not just a measurement. Traversal accuracy is multiplicative in extraction and linking accuracy, whereas lexical and dense retrieval degrade gracefully: partial signal still surfaces something useful. Traversal from a wrong anchor fails *confidently*, emitting a clean typed path — so the interpretability benefit inverts into a liability when the graph or the anchor is wrong. A returned path should be treated as a claim to be checked, not as proof. We think this is the most important practical caveat in the paper.

Where the sheaf idea might still live. Section 8.5 shows the residual fails at path *discrimination* while retaining some ability to detect wrong relation labels and wrong endpoints. That is a verification question — given a claimed triple or chain, is it consistent with the graph? — and it requires neither extraction nor entity linking. It is where we would look next, with neural stalks, and without assuming the answer.

Limitations. Single-node; no clustering, sharding or replication. Pure Python; a Rust hot-path core is unstarted. No dense-retrieval baseline. Evaluated on one external dataset; HotpotQA and MuSiQue loaders are unwritten. The LLM extractor and LLM linker are implemented but unmeasured. Both exploratory layers are unsupported by evidence, one of them negatively so.

10 Conclusion

Knowledge has direction, and metric-ranked retrieval cannot represent direction. That observation is sound, and typed traversal exploits it: under oracle entity linking, CookiX reaches hits@10 0.580 against BM25’s 0.386 on 2WikiMultiHopQA and recovers the gold relation chain on most answered questions. But an end-to-end system must find its own anchor and build its own graph, and there CookiX reaches parity with a lexical baseline from 1994 rather than beating it. Extraction and linking, not traversal, are what stand between those two numbers.

Of the two exotic layers, one has never moved a metric and the other now carries a measured negative result with a structural explanation for it. We report this because a hypothesis that survives only by not being tested is not worth holding. What remains is narrower and, we think, more useful: typed relational retrieval is a real complement to vector search for a specific class of question, it can return the path that justifies an answer, and the honest frontier is upstream — in turning text into edges, and questions into anchors.

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